PNEUMONIA DETECTION FROM CHEST X-RAY IMAGES

Using Deep Learning with MobileNetV2, CBAM Attention & Focal Loss

AI Lab Project Report

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Course	Artificial Intelligence Lab
Topic	Pneumonia Detection from Chest X-Ray Images
Model	MobileNetV2 + CBAM Attention + Focal Loss
Platform	Kaggle GPU (Tesla P100) + Streamlit
Dataset	Chest X-Ray Images (Pneumonia) - Kaggle

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1. Abstract

Pneumonia is a leading cause of mortality worldwide, and timely diagnosis is critical for effective treatment. This project presents an automated pneumonia detection system using deep learning applied to chest X-ray images. We trained a **MobileNetV2** convolutional neural network with transfer learning on a dataset of 5,856 pediatric chest X-rays. To address class imbalance (73% pneumonia, 27% normal) and improve diagnostic accuracy, we incorporated **Convolutional Block Attention Module (CBAM)** for spatial-channel feature refinement and a **Focal Loss** function that focuses learning on hard-to-classify cases. The model achieved **88.14% accuracy**, **94.30% ROC-AUC**, and **95.13% recall** on the test set. A **Streamlit** web application was developed for real-time inference with Grad-CAM heatmap visualization, making the system accessible for clinical screening scenarios.

2. Introduction

Pneumonia is an infection that inflames the air sacs in one or both lungs, and it remains one of the leading causes of death among children under five years old globally. Early and accurate diagnosis is essential to reduce mortality rates. Chest X-ray (CXR) imaging is the most commonly used diagnostic tool for pneumonia, but manual interpretation by radiologists is time-consuming and subject to inter-observer variability.

Deep learning, particularly Convolutional Neural Networks (CNNs), has demonstrated remarkable success in medical image classification tasks. Transfer learning, where a model pre-trained on a large dataset (e.g., ImageNet) is fine-tuned for a specific task, enables high performance even with limited training data.

Objectives:

- Develop a deep learning model for binary classification of chest X-rays (Normal vs Pneumonia)
- Implement transfer learning using MobileNetV2 with CBAM attention mechanism
- Address class imbalance using Focal Loss and weighted sampling
- Build a Streamlit web application for real-time pneumonia screening
- Compare baseline and improved model performances

3. Literature Review

3.1 Transfer Learning in Medical Imaging

Transfer learning has become the standard approach for medical image classification. Models pre-trained on ImageNet encode general visual features (edges, textures, shapes) that transfer well to medical domains. Kermany et al. (2018) demonstrated that InceptionV3 with transfer learning achieved 92.8% accuracy on OCT image classification, establishing the paradigm for medical AI applications.

3.2 Attention Mechanisms

Attention mechanisms allow neural networks to focus on the most informative features. Woo et al. (2018) proposed the Convolutional Block Attention Module (CBAM), which sequentially applies channel and spatial attention to refine feature maps. In medical imaging, CBAM has been shown to improve lesion localization and classification accuracy by 1-3% across multiple benchmarks.

3.3 Focal Loss for Class Imbalance

Lin et al. (2017) introduced Focal Loss to address extreme class imbalance in object detection. The loss function down-weights well-classified examples and focuses training on hard negatives. In pneumonia detection, where datasets are typically imbalanced (70-80% pneumonia cases), Focal Loss has been shown to improve recall by 2-5% compared to standard cross-entropy loss.

3.4 State-of-the-Art Comparison

Model	Dataset	Accuracy	AUC	Year
MobileNetV2 (Baseline)	Chest X-Ray Kaggle	87.98%	0.9582	2026
MobileNetV2 + CBAM + Focal	Chest X-Ray Kaggle	88.14%	0.9430	2026
EfficientNet-B0	NIH Chest X-Ray	89.00%	0.906	2026
DenseNet-121	Chest X-Ray Kaggle	90.00%	-	2025
HybridCATNet	Multi-institutional	96.93%	0.9953	2026

Table 1: Comparison with existing approaches in literature

4. Dataset

The dataset used is the **Chest X-Ray Images (Pneumonia)** dataset by Paul Timothy Mooney, hosted on Kaggle. It contains 5,856 anterior-posterior chest X-ray images in JPEG format, collected from retrospective cohorts of pediatric patients aged 1-5 years at the Guangzhou Women and Children's Medical Center.

4.1 Dataset Distribution

Split	Normal	Pneumonia	Total
Training	1,341	3,875	5,216
Validation	8	8	16
Test	234	390	624
Total	1,583	4,273	5,856

Table 2: Dataset distribution across train/validation/test splits

4.2 Class Imbalance

The dataset exhibits significant class imbalance with approximately **73% Pneumonia** and **27% Normal** cases. This imbalance was addressed through: (1) Focal Loss to down-weight easy examples, (2) class-weighted loss function, and (3) data augmentation to increase effective dataset diversity.

4.3 Data Preprocessing

- Images resized to 224x224 pixels (MobileNetV2 input size)
- Pixel values normalized to [0, 1] range (rescale=1/255)
- RGB channel conversion for consistency
- Training augmentation: rotation (20), shifts (15%), zoom (15%), shear (10%), horizontal flip
- Validation/Test: only rescaling applied

5. Methodology

5.1 Model Architecture

The model uses **MobileNetV2** as the backbone, a lightweight CNN architecture optimized for mobile and embedded devices. MobileNetV2 uses depthwise separable convolutions to reduce parameters while maintaining accuracy. The architecture incorporates **inverted residual blocks** with linear bottlenecks.

Component	Details	Parameters
Backbone	MobileNetV2 (ImageNet pretrained)	3,400,000
CBAM Attention	Channel (r=16) + Spatial (k=7)	~50,000
Global Average Pooling	Reduces spatial dimensions	0
Batch Normalization	Stabilizes training	2,048
Dense Layer 1	256 units + ReLU	328,192
Dropout (0.5 + 0.3)	Regularization	0
Output Layer	1 unit + Sigmoid	257

Table 3: Model architecture components

5.2 CBAM Attention Module

CBAM (Convolutional Block Attention Module) sequentially applies two attention sub-modules: **Channel Attention** and **Spatial Attention**. Channel attention uses both average-pooling and max-pooling to aggregate spatial information, producing a channel attention map. Spatial attention then uses channel-wise average and max pooling to generate a spatial attention map, refining where the model focuses within the feature maps.

5.3 Focal Loss

Standard binary cross-entropy treats all examples equally, which is problematic for imbalanced datasets. Focal Loss introduces a modulating factor that reduces the loss contribution from easy examples and focuses training on hard, misclassified cases:

$$\mathbf{FL(p_t) = -\alpha_t(1 - p_t)^{\gamma} \log(p_t)}$$

We used γ **= 2.0** and α **= 0.25**, following best practices from the literature. This down-weights well-classified examples by a factor of $(1\text{-}p_t)^2$, effectively doubling the focus on hard examples.

5.4 Training Strategy

A two-phase training strategy was employed to maximize performance while preventing overfitting:

Phase	Description	Epochs	Learning Rate
Phase 1	Frozen backbone, train classifier + CBAM	12	1e-4
Phase 2	Unfreeze last 113 layers, fine-tune all	15	1e-5

Table 4: Two-phase training configuration

Additional training configurations: **Adam optimizer**, **ReduceLROnPlateau** scheduler (factor=0.5, patience=2), **Early Stopping** (patience=7, monitor=val_auc, mode=max), and **ModelCheckpoint** saving the best model by validation AUC.

6. Implementation

6.1 Technology Stack

Component	Technology	Version
Deep Learning Framework	TensorFlow / Keras	2.12+
Programming Language	Python	3.10
Web Framework	Streamlit	1.28
Image Processing	OpenCV	4.5+
ML Metrics	scikit-learn	1.0+
Visualization	Matplotlib, Seaborn	3.5+
Compute Platform	Kaggle GPU	Tesla P100
OS	Kaggle Linux / Windows	-

Table 5: Technology stack

6.2 Project Structure

```
Project/
  app.py                  # Streamlit web app
  requirements.txt      # Dependencies
  notebooks/
    pneumonia_train.ipynb    # Baseline model
    pneumonia_improved.ipynb # Improved model
  models/
    mobilenet_pneumonia.h5  # Best model (87.98%)
    baseline_results.json  # Metrics
    improved/             # Improved model files
```

6.3 Key Implementation Details

CBAM Module Implementation: Custom Keras layers for Channel Attention and Spatial Attention were implemented. Channel Attention uses both global average and max pooling with a shared MLP (Dense → ReLU → Dense → Sigmoid). Spatial Attention uses a 7x7 convolution over concatenated channel-wise pooling outputs.

Focal Loss Implementation: Custom loss function implemented using TensorFlow backend operations. Binary cross-entropy is computed per-sample, then multiplied by the modulating factor $(1-p_t)^{\gamma}$ to focus on hard examples.

Grad-CAM Visualization: Gradient-weighted Class Activation Mapping generates heatmaps showing which regions of the X-ray image most influenced the model's decision. This is crucial for interpretability in medical AI applications.

7. Results & Discussion

7.1 Training Results

The baseline MobileNetV2 model was trained for 25 epochs across two phases. Training completed in approximately 25 minutes on a Kaggle Tesla P100 GPU. The model showed stable convergence with no significant overfitting, validated by consistent improvement in validation metrics.

7.2 Model Comparison

Metric	Baseline	Improved	Change
Accuracy	87.98%	88.14%	+0.16%
ROC-AUC	95.82%	94.30%	-1.52%
Precision	87.23%	87.09%	-0.14%
Recall	94.62%	95.13%	+0.51%
F1-Score	90.77%	90.93%	+0.16%

Table 6: Baseline vs Improved model comparison on test set

7.3 Classification Report (Baseline)

Class	Precision	Recall	F1-Score	Support
NORMAL	0.90	0.77	0.83	234
PNEUMONIA	0.87	0.95	0.91	390
Accuracy	-	-	0.88	624
Macro Avg	0.88	0.86	0.87	624
Weighted Avg	0.88	0.88	0.88	624

Table 7: Classification report for the baseline model

7.4 Discussion

Recall Priority: In medical diagnosis, missing a positive case (false negative) is more dangerous than a false alarm (false positive). The baseline model achieves **95% recall for pneumonia**, meaning it correctly identifies 95 out of every 100 pneumonia cases. The improved model further increases this to 95.13%, reducing missed diagnoses.

CLAHE Preprocessing: Contrast Limited Adaptive Histogram Equalization (CLAHE) was tested but significantly degraded performance (66.99% accuracy). This is likely because CLAHE over-normalizes X-ray images, destroying subtle texture differences between normal and pneumonia cases. The original X-ray images already contain sufficient

contrast information for the model.

CBAM + Focal Loss: The combination of CBAM attention and Focal Loss provided marginal improvement in accuracy (+0.16%) and recall (+0.51%). The CBAM module helps the model focus on relevant lung regions, while Focal Loss ensures the model learns from difficult cases.

ROC-AUC Trade-off: The improved model shows a slight decrease in ROC-AUC (-1.52%). This suggests the improvements optimized for point-prediction accuracy at the expense of ranking quality. For deployment, the baseline model may be preferred for screening where threshold-independent evaluation matters.

8. Streamlit Web Application

A **Streamlit** web application was developed to provide an intuitive interface for pneumonia detection. The application features a clean gray-and-white design with real-time prediction and Grad-CAM visualization.

8.1 Features

- Drag-and-drop X-ray image upload (JPG, PNG, BMP)
- Real-time pneumonia detection with confidence score
- Grad-CAM heatmap showing model attention regions
- Overlay visualization combining X-ray and heatmap
- Responsive gray-and-white professional design
- Sidebar with model information and instructions

8.2 Running the Application

```
pip install -r requirements.txt
streamlit run app.py
```

9. Conclusion & Future Work

9.1 Conclusion

This project successfully developed a deep learning-based pneumonia detection system using MobileNetV2 with transfer learning. The key contributions include:

- Achieved **88.14% accuracy** and **95.13% recall** on chest X-ray classification
- Implemented CBAM attention for improved feature focus in medical imaging
- Applied Focal Loss to handle class imbalance in the training dataset
- Built a production-ready Streamlit web application with Grad-CAM visualization
- Demonstrated that recall-focused optimization is critical for medical diagnosis

9.2 Limitations

- Dataset limited to pediatric patients (ages 1-5), may not generalize to adults
- Binary classification only (Normal vs Pneumonia), does not distinguish bacterial vs viral
- Single-institution dataset from Guangzhou, limited geographic diversity
- Validation set is very small (16 images), limiting reliable validation estimates

9.3 Future Work

- Multi-class classification: Normal, Bacterial Pneumonia, Viral Pneumonia
- Incorporate attention-based vision transformers (ViT) for improved accuracy
- Train on larger, multi-institutional datasets (RSNA, CheXpert) for better generalization
- Add lung segmentation as a preprocessing step to focus on relevant regions
- Implement uncertainty estimation for clinical confidence scoring
- Deploy as a cloud-based API for integration with hospital PACS systems

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